# Math for CS & AI: Homework 1

**Instructor:** Kaifeng Lyu (IIIS, Tsinghua University)

(Adapted from earlier versions of this course taught by Andrew Yao)

**Due Date: Mar 7, 2026**

## Reading Task

- Read Chapters 1 and 2 of the following book (available on the course website):
  *Discrete Mathematics: Elementary and Beyond*, by Lovász, Pelikán, and Vesztergombi.

## Problem Set

- Exercises 1–9 from Chapter 2 of the lecture notes:
  https://cloud.tsinghua.edu.cn/d/57ddec714cf74c1cab85/

## Writing Guidelines

- You must write your solutions in LaTeX and in English. Submit your solutions as a single PDF file.
- You may use the following LaTeX template if you don't have your own:
  https://www.overleaf.com/read/pnbrrzmshfrh#f47d73
  You can either (1) download the files and compile them locally, or (2) save a copy on Overleaf (you need to register an account first).
- Check the Writing Guide PDF on the course website. Failing to follow the writing guidelines will result in a penalty of up to 3 points.
- For the first homework, no points will be deducted for writing. However, we may leave comments if your writing could have led to point deductions.
- Need a "math formula" for English writing? Check the following book (available on the course website):
  *Writing Mathematical Papers in English: A Practical Guide*, by Jerzy Trzeciak.